ICS/OT/IoT Threat Model

ABB 2600T Series Model 266 WirelessHART Pressure Transmitter (266DSH WirelessHART variant; OI/266/WIHART-EN Rev. C, 2019)

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> Frameworks: MITRE EMB3D - MITRE ATT&CK for ICS - ISA/IEC 62443 (3-3, 4-2, 4-1, 2-1) - ISA-TR84.00.09 - Attify IoT. Identifiers marked * are model-supplied (not from the embedded reference set).

Executive Summary

The ABB 266 WirelessHART is a battery-powered, microprocessor-based field pressure transmitter (DP/gauge/absolute/flow/level) that communicates over WirelessHART (IEC 62591) to a gateway, with a wired FSK HART maintenance port, local LCD/TTG HMI with keypad, and external non-intrusive Zero/Span/Write-Protect push buttons. Its dominant risk stems from documented, factory-default network-join credentials: Network ID 0xABB (2747) and the AES-128 Join Key hex 57495245 4C455353 4649454C 444B4559 ("WIRELESSFIELDKEY"), which are hardcoded/documented and, if left unchanged, allow an attacker with RF proximity to join the mesh, spoof telemetry, or send unauthorized configuration. The WirelessHART link, while AES-128 authenticated/encrypted per spec, is exposed to wireless sniffing, replay, and rogue-master attacks and can be jammed to deny process visibility. The device exposes a rich configuration surface (DTM over gateway Ethernet, EDD via HART handheld, local HMI) including Simulation mode (C160.030) and Modify Parameter (LRV/URV, burst config), which can falsify the operator's view or manipulate a derived control action. Physical exposure is significant: field-accessible electronics with SW1/SW2 dip-switches (replace/auto-config), Zero/Span buttons, a mechanical write-protect switch, non-volatile config memory, and an internal radio module (FCC ID 2AC5O-WICB) that is susceptible to teardown, bus sniffing, and firmware extraction; no secure-boot, signed-firmware, JTAG/UART protection, or logging details are stated in the manual. Because the 2600T family is TÜV-certified SIL2/SIL3 and the manual states the device may form part of a safety chain, cyber manipulation of the pressure reading is a credible demand source that can also degrade an independent protection layer, so ISA-TR84.00.09 hazard analysis applies. Priority mitigations: change default Network ID/Join Key to unique per-device values, enable hardware write protection, restrict/monitor DTM-over-Ethernet access to the gateway, harden the physical enclosure/dip-switches, and treat the transmitter's reading as untrusted input in any safety/control decision.

1. System Model

Field-mounted WirelessHART differential/gauge/absolute pressure transmitter measuring process pressure and transmitting digital telemetry over a WirelessHART mesh to a gateway. The gateway bridges to plant systems via Ethernet TCP/IP (Asset Vision Basic, DTM) and to a DCS via Modbus RS-485. Local configuration is available via LCD/TTG HMI keypad, external Zero/Span/Write-Protect push buttons, and a wired FSK HART maintenance port for handheld/modem. The transmitter joins the wireless network using a shared Network ID and 128-bit Join Key.

Components

Component	Role	Sub-components
Pressure Sensor / Process Interface	Capacitive/piezoresistive DP sensor and flanges measuring process pressure; sensor NV memory stores flange/O-ring/fill-fluid data	Sensor element, Sensor NV memory, Process flanges/manifold
Electronic/Communication Board (MCU)	Microprocessor board running measurement, HART 7 stack, HMI, diagnostics; hosts LCD connection and config NV memory	Microprocessor/MCU, Non-volatile config memory, SW1/SW2 dip-switches (replace/auto-config), LCD/communication interface
WirelessHART Radio Module (WICB)	2.4 GHz IEEE 802.15.4 WirelessHART transceiver; forms/joins mesh, performs AES-128 auth/encryption	RF SoC/transceiver, External rotatable antenna, Radio firmware
Local HMI	LCD or TTG display with 4-button keypad for menu configuration and Device Info; supports display password	LCD/TTG display, Keypad, Zero/Span/Write-Protect push buttons
Power Module	3.6V D-cell lithium-thionyl-chloride battery (Tadiran SL-2880 / ABB DR3051), field-serviceable	Lithium battery, Battery holder/terminals

Component	Role	Sub-components
WirelessHART Gateway (adjacent system)	Network Manager; bridges mesh to Ethernet TCP/IP and Modbus RS-485 to DCS; holds Network ID/Join Key	Network Manager, Ethernet interface, RS-485 interface, Advertising/join control

Interfaces

Interface	Type	Protocol	Exposure
WirelessHART mesh RF	wireless	WirelessHART / IEC 62591 (HART 7 over IEEE 802.15.4, AES-128)	external
Wired FSK HART maintenance port	physical	HART FSK (Bell 202)	physical
Local HMI keypad / TTG	physical	Menu UI	physical
External Zero/Span/Write-Protect push buttons	physical	Magnetic/non-intrusive button	physical
SW1/SW2 dip-switches	physical	Board jumper/switch (replace/auto-config mode)	physical
Internal radio-to-MCU link	bus	Internal serial/SPI (unspecified)	internal
DTM over Gateway Ethernet	api	HART-IP/DTM over Ethernet TCP/IP (via gateway)	zone

Data Flows

From	To	Data	Protocol	Crosses boundary
Pressure Sensor / Process Interface	Electronic/Communication Board (MCU)	Raw pressure/temperature measurement	Internal analog/digital	No
Electronic/Communication Board (MCU)	WirelessHART Gateway (adjacent system)	Process variable telemetry, burst messages, status, diagnostics	WirelessHART (AES-128)	Yes
WirelessHART Gateway (adjacent system)	Electronic/Communication Board (MCU)	Join grant, config/write commands, burst setup	WirelessHART (AES-128)	Yes
HART Handheld/Modem	Electronic/Communication Board (MCU)	EDD-based config, calibration, Network ID/Join Key write, simulation	HART FSK	Yes
Asset Vision Basic / DTM (PC)	WirelessHART Gateway (adjacent system)	Remote config/diagnostics of device	Ethernet TCP/IP	Yes
WirelessHART Gateway (adjacent system)	DCS	Process variable to control/monitoring	Modbus RS-485	Yes

2. Device Model - EMB3D Properties

Property	Category	Feature / Interface	Present	Notes
PID-11	Hardware	Microprocessor-based electronic/communication board	Yes	Manual describes microprocessor-based transmitter; specific MCU/SoC not stated.
PID-12	Hardware	Non-volatile config memory on electronic board and sensor	Yes	NV memory stores LRV/URV, flange/O-ring/fill-fluid data; burn errors M140.024/M145.018.
PID-123	Hardware	ROM/NVRAM for firmware and parameters	Yes	Config persisted ~10s after button press; auto-config restores params after board/sensor swap.
PID-13	Hardware	Peripheral chips: radio module, LCD, sensor over internal bus	Yes	Radio (WICB) communicates internally with electronics (F210.015 Radio Failure).
PID-14	Hardware	Wired FSK HART maintenance terminals	Yes	External serial-type maintenance port for handheld/modem.

Property	Category	Feature / Interface	Present	Notes
PID-15	Hardware	Hardware access port (JTAG/UART) - unconfirmed	Yes	No JTAG/UART/SWD documented; likely present for manufacturing; treated as assumption.
PID-21	System Software	Bootloader (implied)	Yes	Firmware/HW/SW revisions exposed via EDD/HMI; boot chain not documented.
PID-26	System Software	Firmware update support unclear	Yes	No documented field firmware update path; in-field protocol upgrade referenced for family. Update authenticity unknown.
PID-271	System Software	Firmware integrity validation not documented	Yes	No signed/secure-boot claims in manual; assume lacking cryptographic integrity validation.
PID-28	System Software	Event/diagnostic states (NAMUR NE107)	Yes	Failure/Maintenance/Out-of-Spec/Function-Check categories, error codes; no security audit log documented.
PID-31	Application Software	Measurement/HART application, HMI menus, Simulation	Yes	Simulation mode C160.030 overrides PdP/static/temp/output; Squawk, Load Factory Settings, Lock/Unlock.
PID-3321	Application Software	Display protection password ('security')	Yes	Optional display menu password; strength/policy not specified.
PID-331	Application Software	Unauthenticated local config via Zero/Span/HMI	Yes	Local buttons/HMI allow config unless write-protect enabled; no user auth for local ops.
PID-41	Networking	WirelessHART remote services (join, burst, HART cmds)	Yes	Supports HART cmds #1,#2,#3,#9,#33,#48,#73; burst broadcast; joins mesh via Network ID + Join Key.
PID-411	Networking	Remote config modification over WirelessHART/DTM	Yes	Config, calibration, parameters writable remotely via gateway/DTM.
PID-4113	Networking	WirelessHART AES-128 auth/encryption	Yes	Join Key drives AES-128 network security per HART spec; default key is documented/hardcoded.
PID-42	Networking	Mesh routing/forwarding	Yes	WirelessHART devices route messages for other nodes in the mesh.

3. EMB3D Threat -> Mitigation Table

Element	TID	Threat	Lik.	Imp.	Mitigations	Summary
WirelessHART join credentials (Network ID 0xABB + Join Key WIRELESSFIELDKEY)	TID-317	Predictable Cryptographic Key	High	High	MID-033, MID-047, MID-043	Provision unique, unpredictable per-device Network ID/Join Key; manage/rotate defaults; support individual join keys.
WirelessHART join / documented default key	TID-328	Hardcoded Credentials	High	High	MID-043, MID-033	Force change of default join credentials at commissioning; forbid shared defaults across networks.
WirelessHART RF link	TID-407	Missing Message Replay Protection	Medium	High	MID-034, MID-036, MID-037	Enforce nonces and network timestamps; validate freshness per WirelessHART security.
WirelessHART RF link	TID-406	Unauthorized Messages or Connections	Medium	High	MID-034, MID-018, MID-038	Authenticate all commands; require authenticated/authorized writes for config changes.
WirelessHART protocol/PHY	TID-404	Remotely Triggerable Deadlock/DoS	Medium	Medium	MID-080, MID-032	Channel hopping resilience, request limits, and out-of-band monitoring; treat telemetry loss as safe-state input.

Element	TID	Threat	Lik.	Imp.	Mitigations	Summary
WirelessHART traffic confidentiality	TID-408	Unencrypted Sensitive Data Communication	Low	Medium	MID-035, MID-044	Enforce AES-128 encryption end-to-end; avoid weak/downgrade configurations.
WirelessHART crypto protocol	TID-411	Weak/Insecure Cryptographic Protocol	Medium	High	MID-044, MID-033, MID-028	Use individual join keys, strong algorithms, and hardware-backed key storage.
Firmware / radio module	TID-115	Firmware/Data Extraction via Hardware Interface	Medium	High	MID-057, MID-058, MID-054	Disable/lock debug ports; enable readout protection; encrypt NV storage.
Bootloader / firmware	TID-211	Device Allows Unauthenticated Firmware Installation	Low	High	MID-026, MID-002, MID-029	Implement secure, authenticated firmware update and hardware root of trust.
Non-volatile config memory	TID-108	ROM/NVRAM Data Extraction or Modification	Medium	High	MID-054, MID-052, MID-031	Encrypt/authenticate NV contents; protect board traces; require physical presence for changes.
SW1/SW2 dip-switches & auto-config	TID-116	Latent Privileged Access Port	Medium	Medium	MID-031, MID-016	Physical presence controls, tamper-evident enclosure, restrict maintenance access.
Application Simulation mode (C160.030)	TID-224	Excessive Access via Software Diagnostic Features	Medium	High	MID-039, MID-018, MID-017	Restrict/authenticate diagnostic & simulation functions; alarm/log when active.
Local HMI / Zero-Span buttons	TID-311	Default Credentials	Medium	Medium	MID-043, MID-018, MID-031	Set strong display password; engage hardware write protection; enforce physical presence.
Diagnostic/event logging	TID-225	Logs can be manipulated on the device	Medium	Medium	MID-017, MID-085	Provide security-relevant auditing and export logs off device to gateway/historian.
DTM-over-Ethernet remote access	TID-303	Excessive Trust in Offboard Management/IDE Software	Medium	High	MID-018, MID-038, MID-081	Authenticate administrative actions, secure tunnels for DTM, and harden/segment engineering hosts.

4. MITRE ATT&CK for ICS

Tactic	Technique ID	Technique	Applicability	Related EMB3D
Initial Access	T0860	Wireless Compromise	Attacker in RF range uses default/known Network ID+Join Key to join the WirelessHART mesh.	TID-317, TID-328, TID-406
Initial Access	T0822	External Remote Services	DTM/Asset Vision Basic reaches the device via gateway over intranet Ethernet, an exploitable remote path.	TID-303, TID-411
Discovery	T0887	Wireless Sniffing	Sniff 2.4 GHz WirelessHART traffic to enumerate devices, Long-TAGs, and network topology.	TID-408, TID-407
Discovery	T0888	Remote System Information Discovery	HART cmd #73 Find Device and Device Info reveal HW/SW/device revisions and identity.	TID-224
Collection	T0801	Monitor Process State	Read burst PV/telemetry to learn process pressure and time manipulation.	TID-408
Command and Control	T0869	Standard Application Layer Protocol	Use standard WirelessHART/HART commands as the control channel to blend in.	TID-406, TID-401

Tactic	Technique ID	Technique	Applicability	Related EMB3D
Inhibit Response Function	T0814	Denial of Service	RF jamming or malformed traffic denies telemetry, causing Loss of View of pressure.	TID-404
Inhibit Response Function	T0878	Alarm Suppression	Modify alarm/limit settings or spoof normal status to suppress abnormal-pressure alarms.	TID-224, TID-406
Impair Process Control	T0836	Modify Parameter	Alter LRV/URV, calibration, burst config to distort the transmitted process value.	TID-406, TID-224
Impair Process Control	T0855	Unauthorized Command Message	Send unauthorized HART write/simulation commands over the joined mesh.	TID-406, TID-407
Impair Process Control	T0831	Manipulation of Control	A falsified pressure PV drives an incorrect control action in the DCS loop that consumes the reading.	TID-224, TID-836
Evasion	T0856	Spoof Reporting Message	Inject spoofed burst/telemetry so operators see falsified pressure while process deviates.	TID-407, TID-224
Evasion	T0830	Adversary-in-the-Middle	Rogue node/master intercepts and alters mesh messages between transmitter and gateway.	TID-406, TID-407
Persistence	T0839	Module Firmware	Implant modified radio/electronic-module firmware via physical/debug access for persistence.	TID-115, TID-211
Impact	T0880	Loss of Safety	Manipulated/denied pressure reading defeats a protection layer relying on the transmitter, risking a hazardous condition.	TID-224, TID-406, TID-404

5. Attify IoT Reverse-Engineering / Attack Surface

Domain	Vector	Technique	Finding / Hypothesis	Related EMB3D
Hardware / Device Teardown	Debug/test interface discovery (UART/JTAG/SWD)	Open enclosure, probe electronic/communication board, JTAGulator/logic analyzer to find manufacturing debug port and open a shell.	No debug ports documented but likely present for manufacturing; if unlocked, allows firmware/key extraction.	TID-115, TID-119
Hardware / Device Teardown	Chip-off / flash extraction	Desolder NV flash/EEPROM on electronic and sensor boards; read with CH341A/flashrom to dump config and keys.	Config NV stores LRV/URV, calibration, and potentially join credentials; unencrypted NV assumed.	TID-108, TID-111
Hardware / Device Teardown	Physical tamper / port abuse (SW1/SW2)	Toggle SW1/SW2 dip-switches into replace/auto-config; use Zero/Span buttons to rerange without authentication.	Field-accessible switches/buttons enable reconfiguration; write-protect mitigates only if engaged.	TID-116, TID-311
Firmware Analysis	Secret & credential hunting	binwalk/entropy scan of dumped firmware to locate hardcoded Join Key handling and default Network ID.	Default key 'WIRELESSFIELDKEY' and Network ID 0xABB are documented; firmware may embed default handling.	TID-328, TID-317, TID-226

Domain	Vector	Technique	Finding / Hypothesis	Related EMB3D
Firmware Analysis	Update/boot chain analysis	Inspect for signing/secure boot and rollback checks; assess firmware update authenticity.	No secure-boot/signed-firmware claims in manual; unauthenticated firmware install plausible.	TID-211, TID-201
Radio / Wireless Communication	Sub-GHz/2.4GHz proprietary RF (WirelessHART)	IEEE 802.15.4/WirelessHART sniffer + SDR to capture join/advertise traffic; attempt join with default credentials; replay frames.	Default Network ID+Join Key allow join; replay feasible if freshness weak; jamming denies telemetry.	TID-406, TID-407, TID-404, TID-411
Network Services	ICS/OT protocol testing (HART commands)	Issue HART cmds #1/#2/#3/#9/#33/#48/#73 and write/simulation commands over the joined mesh; fuzz parsers.	Unauthorized command/parameter modification and simulation enable PV falsification.	TID-406, TID-224, TID-401
Network Services	Cleartext & MitM via gateway/DTM	Access DTM over gateway Ethernet from a compromised engineering host; adversary-in-the-middle on mesh.	DTM-over-Ethernet remote path over-trusts engineering PC; reconfiguration possible.	TID-303, TID-406
Application	Local HMI / display password & write-protect	Attempt default/blank display password; verify write-protect state; use Load Factory Settings/Reset.	Weak/optional display password and unauthenticated local config where write-protect not engaged.	TID-311, TID-224

6. ISA/IEC 62443 Requirements Mapping

6.1 62443-4-2 Component Requirements

Req ID	Name	Design Mitigation	Addresses
CR 1.6 / NDR 1.6	Wireless access management	Authenticate/authorize each device joining the WirelessHART mesh with unique credentials; forbid shared/default Network ID+Join Key.	TID-317, TID-328, TID-406
CR 1.5	Authenticator management	Eliminate hardcoded/default Join Key; force change at commissioning; support individual per-device join keys and secure lifecycle.	TID-328, TID-317, TID-411
CR 1.14	Strength of symmetric key authentication	Protect and validate the AES-128 symmetric join/session keys with adequate entropy and hardware-backed storage.	TID-411, TID-317
CR 2.1	Authorization enforcement	Enforce least-privilege RBAC so only authorized roles can write parameters, run simulation, or change alarm/burst settings.	TID-224, TID-406
CR 2.2 / NDR 1.6	Wireless use control	Authorize, monitor, and restrict wireless usage; alarm on unexpected joins and default-credential use.	TID-406, TID-328
CR 3.1	Communication integrity	Ensure WirelessHART message authentication with anti-replay (nonces/timestamps) for all telemetry and commands.	TID-407, TID-406
CR 3.5	Input validation	Validate HART command syntax/content and join parameters to resist malformed/fuzzed traffic.	TID-404, TID-401
CR 3.6 / EDR 3.6	Deterministic output	On loss of comms or detected fault, report defined status/NE107 category so consumers apply a safe/known state.	TID-404

Req ID	Name	Design Mitigation	Addresses
CR 3.10 / EDR 3.10	Support for updates	Provide authenticated, integrity-verified firmware update with rollback protection.	TID-211, TID-201
EDR 3.11	Physical tamper resistance and detection	Add tamper-evident enclosure/switch protection; detect and log opening and dip-switch/button changes.	TID-116, TID-108, TID-311
CR 2.13 / EDR 2.13	Use of physical diagnostic and test interfaces	Disable or lock manufacturing debug/test ports; require authentication for the wired FSK maintenance port.	TID-115, TID-119
CR 4.1	Information confidentiality	Protect join keys and calibration secrets at rest (encrypted NV) and in transit (AES-128 mesh).	TID-108, TID-408
CR 2.8	Auditable events	Generate security audit records for joins, writes, simulation enable, and write-protect changes.	TID-224, TID-225
CR 7.1	Denial of service protection	Maintain essential measurement/status reporting resilience under RF jamming/flooding; mesh path diversity.	TID-404

6.2 62443-3-3 System Requirements

Req ID	Name	System Control
SR 1.6	Wireless access management	Manage and authorize all wireless devices/users; commission unique Network ID/Join Key per network and rotate defaults.
SR 2.2	Wireless use control	Authorize, monitor, and enforce usage restrictions on WirelessHART; detect rogue joins/masters.
SR 1.5	Authenticator management	Prohibit default/hardcoded credentials at the system level; manage join key lifecycle in the gateway/network manager.
SR 3.1	Communication integrity	Enforce authenticated, anti-replay comms across the mesh and gateway conduits.
SR 5.1	Network segmentation	Segment the WirelessHART/gateway network from plant IT and the DCS; place engineering DTM access behind controlled conduits.
SR 5.2	Zone boundary protection	Deny-by-default at the gateway Ethernet/Modbus boundary; restrict DTM/Asset Vision Basic reachability.
SR 7.1	Denial of service protection	Protect essential monitoring functions from RF DoS; provide out-of-band or redundant measurement where safety-relevant.
SR 6.2	Continuous monitoring	Monitor WirelessHART diagnostics (Radio Failure, Join Failure) and anomalous PV/telemetry for tamper detection.
SR 3.6	Deterministic output	Ensure consuming systems treat lost/stale WirelessHART telemetry as a defined safe/known state.

6.3 62443-4-1 Secure Development Lifecycle

Practice	Name	Recommendation
SR-2	Specification of Security Requirements / Threat Model	Maintain a product threat model covering default-credential risk, wireless attack surface, and physical field exposure.
SD	Secure by Design	Remove documented default Join Key; require per-device unique credentials; attack-surface reduction on diagnostic/simulation and debug ports.
SI	Secure Implementation	Apply secure coding to the HART parser and join logic; hardware-backed key storage for AES-128 keys.
SVV	Security Verification and Validation Testing	Fuzz WirelessHART/HART interfaces, test replay/jamming resilience, and pen-test default-credential join scenarios.
SUM	Security Update Management	Provide qualified, authenticated firmware updates and advisories; define patch delivery for field transmitters.

Practice	Name	Recommendation
DM	Management of Security-related Issues	Operate a vulnerability handling process (PSIRT) and publish advisories for the 266 WirelessHART family.
SG	Security Guidelines	Provide hardening guidance: change defaults, engage write-protect, secure DTM/gateway access, decommission key erasure.

7. ISA-TR84.00.09 Safety / Hazard / Consequence

Applicable: Yes - SIS/SIF: The 2600T Model 266 family is TÜV NORD certified for SIL2 (1oo1) and SIL3 (1oo2) service, and the manual states the transmitter may form part of a safety chain. In a SIF, the 266 acts as the pressure SENSOR element; the WirelessHART variant is typically used for monitoring/BPCS but where pressure is a demand input to a protection layer, cyber manipulation of its reading is safety-relevant. Wireless telemetry should not be the sole basis of a high-SIL final protective action; ISA-TR84.00.09 analysis is warranted.

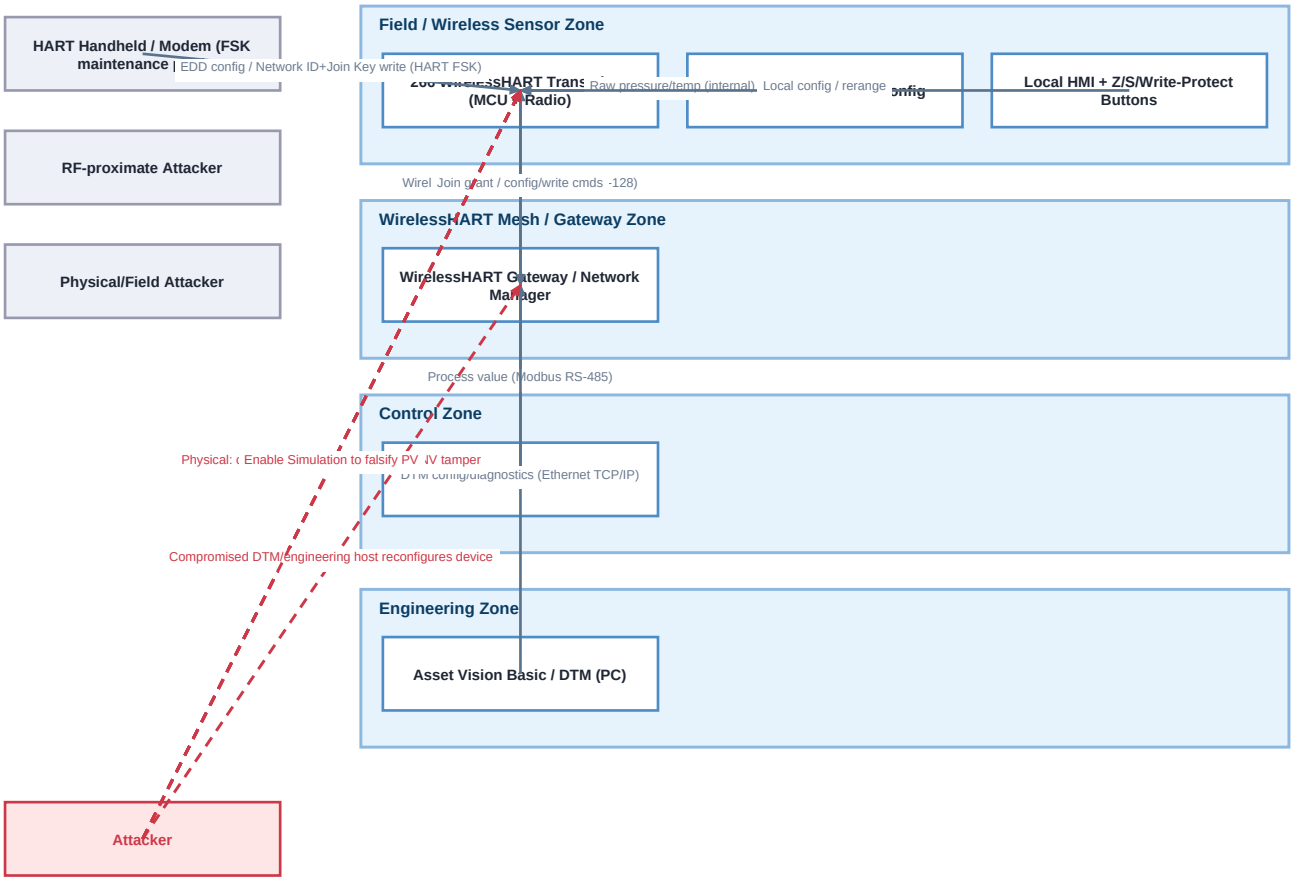
Scenario	Attack Path	Consequence	Severity	Defeats IPLs	Countermeasures
Falsified low/normal pressure while true process pressure rises (overpressure not detected)	T0860 Wireless Compromise (default Network ID+Join Key) -> TID-406 Unauthorized Messages -> T0836 Modify Parameter / TID-224 Simulation (C160.030) -> T0856 Spoof Reporting Message -> T0878 Alarm Suppression -> T0880 Loss of Safety	Overpressure demand undetected; relief/protection not triggered on the sensed signal; potential vessel/line rupture, release, fire.	Major	Yes	Change default join credentials; do not rely on a single wireless transmitter for SIF demand; independent SIS sensor in a separate 62443 zone; alarm when simulation active; deterministic safe-state on stale telemetry; TR84-CRA integrated with HAZOP/LOPA.
Denial of View via RF jamming during a process transient	T0887 Wireless Sniffing (timing) -> T0814 Denial of Service (2.4GHz jam) / TID-404 -> T0815 Denial of View	Operators lose pressure visibility during a critical transient, delaying corrective action.	Serious	Yes	Treat loss of WirelessHART telemetry as a defined safe/known state (SR 3.6/CR 3.6); provide redundant/wired measurement for safety-critical loops; monitor Radio/Join Failure diagnostics.
Persistent PV bias via physical reconfiguration in the field	Physical access -> TID-116 SW1/SW2 replace/auto-config + Zero/Span buttons -> TID-108 NVRAM modification (LRV/URV/calibration) -> T0831 Manipulation of Control	Systematically biased pressure reading drives incorrect control/protection response over time.	Major	Yes	Engage hardware write protection; tamper-evident enclosure and dip-switch guards; MoC (TR84-MOC) and periodic calibration verification; physical presence controls.

8. Risk Register

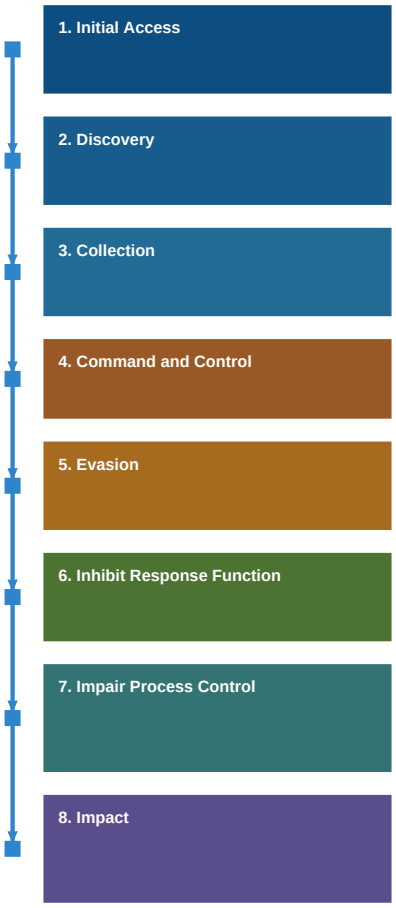
ID	Risk	Threats	ATT&CK	L	I	Level	Recommendation	Requirements
R1	Documented default Network ID (0xABB) and Join Key ('WIRELESSFIELDKEY') left unchanged allow an RF-proximate attacker to join the mesh and spoof/command the transmitter.	TID-317, TID-328, TID-406	T0860, T0855	H	H	Critical	Mandate change of Network ID and Join Key to unique, unpredictable per-device values at commissioning; use individual join keys; verify no default remains.	CR 1.5, CR 1.6, SR 1.5, SR 1.6

ID	Risk	Threats	ATT&CK	L	I	Level	Recommendation	Requirements
R2	Falsification of the pressure PV via Simulation mode / parameter modification leads to spoofed operator view and impaired control/protection.	TID-224, TID-406, TID-407	T0836, T0856, T0878, T0831	M	H	High	Restrict/authenticate simulation and write functions; alarm and log when simulation active; RBAC on parameter writes; monitor for anomalous PV.	CR 2.1, CR 2.8, SR 6.2
R3	RF jamming or malformed traffic denies WirelessHART telemetry, causing Loss of View during process transients.	TID-404	T0814, T0815	M	M	High	Ensure consumers treat lost/stale telemetry as safe-state; provide redundant/wired measurement for safety loops; monitor Radio/Join Failure diagnostics.	CR 3.6, SR 3.6, SR 7.1
R4	Replay/AiTM on the mesh injects stale or altered telemetry/commands if freshness protections are weak.	TID-407, TID-406	T0830, T0856	M	H	High	Enforce nonces/timestamps and message authentication; validate freshness per WirelessHART security; monitor for duplicate/late frames.	CR 3.1, SR 3.1
R5	Physical field access enables firmware/key extraction and NV/config tampering (chip-off, debug ports, SW1/SW2, buttons).	TID-115, TID-108, TID-116, TID-311	T0839, T0831	M	H	High	Engage hardware write protection; lock/disable debug ports; encrypt NV; tamper-evident enclosure; physical access controls and periodic calibration checks.	EDR 2.13, EDR 3.11, CR 4.1
R6	Remote reconfiguration via DTM over gateway Ethernet from a compromised engineering host.	TID-303, TID-406	T0822, T0855	M	H	High	Authenticate administrative actions; secure tunnels for DTM; segment and harden engineering workstations and the gateway boundary (deny-by-default).	CR 2.1, SR 5.1, SR 5.2
R7	Unauthenticated/unverified firmware update path could allow malicious firmware persistence.	TID-211, TID-201	T0839, T0857	L	H	Medium	Implement secure boot, signed firmware, and rollback protection; provide authenticated update mechanism and advisories.	EDR 3.10, EDR 3.14
R8	Weak/optional display password and unauthenticated local reranging where write-protect is not engaged.	TID-311, TID-224	T0823	M	M	Medium	Set strong display password; engage hardware write protection by default policy; require physical presence for local changes.	CR 1.7, CR 2.1

9. Threat Model Diagram



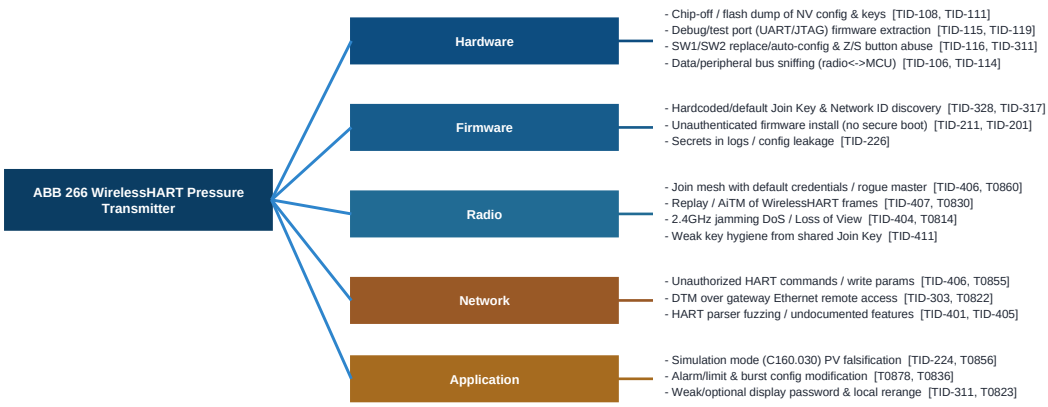
10. ATT&CK for ICS Kill Chain



11. SANS ICS Cyber Kill Chain



12. Attack Mind Map



Assumptions

- No CPU/SoC, memory sizes, JTAG/UART/SWD debug ports, or crypto implementation details are stated in the manual; hardware access ports (PID-15) are assumed present for manufacturing.
- WirelessHART security (AES-128 authentication/encryption keyed by the Join Key) is assumed per the HART7/IEC 62591 specification; the manual only implies it.
- Default Network ID 0xABB (2747) and Join Key 57495245 4C455353 4649454C 444B4559 are treated as factory/documented defaults per the ingested excerpt; actual field configuration may differ.
- No signed-firmware, secure-boot, tamper detection, or security audit logging is documented; assumed absent unless proven otherwise.
- No published CVEs for this specific model were provided; risk is derived from architecture, defaults, and physical

exposure.

- The WirelessHART variant is typically used for monitoring/BPCS; SIL2/SIL3 certification of the 2600T family and the 'safety chain' statement imply possible safety relevance, so ISA-84 is treated as applicable.
- Gateway, Asset Vision Basic, and DCS are adjacent systems included only where they affect the transmitter's attack surface.

References

- ABB 266 WirelessHART Pressure Transmitter Manual OI/266/WIHART-EN Rev. C (2019) - OI_266WIHART-EN-C-10_2019.pdf
- ABB WirelessHART pressure transmitter.pdf (2600T family datasheet: SIL2/SIL3 TÜV NORD, TTG HMI, DTM/HHT, PILD diagnostics)
- EMB3D Threat Model framework (device properties PID, threats TID, mitigations MID)
- MITRE ATT&CK for ICS (tactics/techniques T0xxx)
- IEC 62443-3-3 System Requirements; IEC 62443-4-2 Component Requirements (CR/EDR/NDR); IEC 62443-4-1 SDL practices
- ISA-TR84.00.09 (safety/security integration with IEC 61511/62443)
- HART Communication Foundation WirelessHART / IEC 62591 security (AES-128) - referenced for protocol security assumptions
- Attify IoT reverse-engineering attack-surface methodology